data_processing

May 15, 2026

1 MedQuAD dataset exploration

- Dataset is in `csv` format and contains 4 columns: question, answer, source and focus_area
- The total number of records is 16 412

[1]:
```python
import pandas as pd
ROOT = "../"

# Load CSV
df = pd.read_csv(ROOT + "dataset/medquad.csv")

print(f"Number of rows: {len(df)}")

# Display overall structure
display(df.head(10))
print(df.dtypes)

# Display a few full examples
for i in range(3):
    print(f"Example {i + 1}:")
    print(f"Question: {df.iloc[i]["question"]}")
    print(f"Answer: {df.iloc[i]["answer"]}")
    print(f"Focus Area: {df.iloc[i]["focus_area"]}")
    print("-" * 100)
```

```
Number of rows: 16412

                                    question  \
0                  What is (are) Glaucoma ?
1                    What causes Glaucoma ?
2     What are the symptoms of Glaucoma ?
3  What are the treatments for Glaucoma ?
4                  What is (are) Glaucoma ?
5                  What is (are) Glaucoma ?
6                  What is (are) Glaucoma ?
7            Who is at risk for Glaucoma? ?
8                   How to prevent Glaucoma ?
9     What are the symptoms of Glaucoma ?
```

```
                                                            answer          source  \
0  Glaucoma is a group of diseases that can damag…  NIHSeniorHealth
1  Nearly 2.7 million people have glaucoma, a lea…  NIHSeniorHealth
2  Symptoms of Glaucoma  Glaucoma can develop in …  NIHSeniorHealth
3  Although open-angle glaucoma cannot be cured, …  NIHSeniorHealth
4  Glaucoma is a group of diseases that can damag…  NIHSeniorHealth
5  The optic nerve is a bundle of more than 1 mil…  NIHSeniorHealth
6  Open-angle glaucoma is the most common form of…  NIHSeniorHealth
7  Anyone can develop glaucoma. Some people are a…  NIHSeniorHealth
8  At this time, we do not know how to prevent gl…  NIHSeniorHealth
9  At first, open-angle glaucoma has no symptoms…  NIHSeniorHealth

  focus_area
0   Glaucoma
1   Glaucoma
2   Glaucoma
3   Glaucoma
4   Glaucoma
5   Glaucoma
6   Glaucoma
7   Glaucoma
8   Glaucoma
9   Glaucoma

question      str
answer        str
source        str
focus_area    str
dtype: object
Example 1:
Question: What is (are) Glaucoma ?
Answer: Glaucoma is a group of diseases that can damage the eye's optic nerve
and result in vision loss and blindness. While glaucoma can strike anyone, the
risk is much greater for people over 60. How Glaucoma Develops  There are
several different types of glaucoma. Most of these involve the drainage system
within the eye. At the front of the eye there is a small space called the
anterior chamber. A clear fluid flows through this chamber and bathes and
nourishes the nearby tissues. (Watch the video to learn more about glaucoma. To
enlarge the video, click the brackets in the lower right-hand corner. To reduce
the video, press the Escape (Esc) button on your keyboard.) In glaucoma, for
still unknown reasons, the fluid drains too slowly out of the eye. As the fluid
builds up, the pressure inside the eye rises. Unless this pressure is
controlled, it may cause damage to the optic nerve and other parts of the eye
and result in loss of vision. Open-angle Glaucoma The most common type of
glaucoma is called open-angle glaucoma. In the normal eye, the clear fluid
leaves the anterior chamber at the open angle where the cornea and iris meet.
When fluid reaches the angle, it flows through a spongy meshwork, like a drain,
and leaves the eye. Sometimes, when the fluid reaches the angle, it passes too
```

slowly through the meshwork drain, causing the pressure inside the eye to build. If the pressure damages the optic nerve, open-angle glaucoma -- and vision loss -- may result. There is no cure for glaucoma. Vision lost from the disease cannot be restored. However, there are treatments that may save remaining vision. That is why early diagnosis is important. See this graphic for a quick overview of glaucoma, including how many people it affects, whos at risk, what to do if you have it, and how to learn more. See a glossary of glaucoma terms.
Focus Area: Glaucoma

Example 2:
Question: What causes Glaucoma ?
Answer: Nearly 2.7 million people have glaucoma, a leading cause of blindness in the United States. Although anyone can get glaucoma, some people are at higher risk. They include - African-Americans over age 40 - everyone over age 60, especially Hispanics/Latinos - people with a family history of glaucoma. African-Americans over age 40 everyone over age 60, especially Hispanics/Latinos people with a family history of glaucoma. In addition to age, eye pressure is a risk factor. Whether you develop glaucoma depends on the level of pressure your optic nerve can tolerate without being damaged. This level is different for each person. Thats why a comprehensive dilated eye exam is very important. It can help your eye care professional determine what level of eye pressure is normal for you. Another risk factor for optic nerve damage relates to blood pressure. Thus, it is important to also make sure that your blood pressure is at a proper level for your body by working with your medical doctor. (Watch the animated video to learn more about the causes of glaucoma. To enlarge the video, click the brackets in the lower right-hand corner. To reduce the video, press the Escape (Esc) button on your keyboard.)
Focus Area: Glaucoma

Example 3:
Question: What are the symptoms of Glaucoma ?
Answer: Symptoms of Glaucoma Glaucoma can develop in one or both eyes. The most common type of glaucoma, open-angle glaucoma, has no symptoms at first. It causes no pain, and vision seems normal. Without treatment, people with glaucoma will slowly lose their peripheral, or side vision. They seem to be looking through a tunnel. Over time, straight-ahead vision may decrease until no vision remains. Tests for Glaucoma Glaucoma is detected through a comprehensive eye exam that includes a visual acuity test, visual field test, dilated eye exam, tonometry, and pachymetry. (Watch the animated video to learn more about testing for glaucoma. To enlarge the video, click the brackets in the lower right-hand corner. To reduce the video, press the Escape (Esc) button on your keyboard.) A visual acuity test uses an eye chart test to measure how well you see at various distances. A visual field test measures your side or peripheral vision. It helps your eye care professional tell if you have lost side vision, a sign of glaucoma. In a dilated eye exam, drops are placed in your eyes to widen, or dilate, the pupils. Your eye care professional uses a special magnifying lens to

```
examine your retina and optic nerve for signs of damage and other eye problems.
After the exam, your close-up vision may remain blurred for several hours. In
tonometry, an instrument measures the pressure inside the eye. Numbing drops may
be applied to your eye for this test. With pachymetry,  a numbing drop is
applied to your eye. Your eye care professional uses an ultrasonic wave
instrument to measure the thickness of your cornea.
Focus Area: Glaucoma
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```

2 Filtering the dataset

- Our primary goal is getting the diesease name and its corresponding symptopms. In order to achieve that, we will filter the data to only get the records that either contain any of the specified `keywords` related to symptoms in either the `question` or the `answer` field
- After filtering the data by this logic, we get 14 884 records. There are 4985 unique `focus_areas` labels, and by inspecting a few of them, we can see that they indeed correspond to disease names.

[2]:
```python
import pandas as pd

# Load CSV
df = pd.read_csv(ROOT + "dataset/medquad.csv")

# We only need the data that contains some information about the disease and␣
↪its symptoms
keywords = [
    "symptom",
    "signs",
    "cause",
    "disease",
    "condition",
    "syndrome"
]

# Create regex pattern
pattern = "|".join(keywords)

# Filter the data by checking if any of the keywords is either in the question␣
↪or the answer field
filtered = df[
    df.iloc[:, 0].str.contains(pattern, case=False, na=False) |
    df.iloc[:, 1].str.contains(pattern, case=False, na=False)
]

# Show some filtered results
for i in range(5):
```

```
    print(f"Example {i}:")
    print(f"Question: {filtered.iloc[i]["question"]}")
    print(f"Answer: {filtered.iloc[i]["answer"]}")
    print(f"Focus Area: {filtered.iloc[i]["focus_area"]}")
    print("-" * 100)


print(f"Number of records: {filtered.shape[0]}")
print(f"Number of unique focus_areas: {filtered['focus_area'].nunique()}") #␣
↪This will ideally contain disease names

# Show some unique focus areas
print(f"Example of focus_areas: {filtered["focus_area"].dropna().unique()[:
↪25]}")
```

```
Example 0:
Question: What is (are) Glaucoma ?
Answer: Glaucoma is a group of diseases that can damage the eye's optic nerve
and result in vision loss and blindness. While glaucoma can strike anyone, the
risk is much greater for people over 60. How Glaucoma Develops  There are
several different types of glaucoma. Most of these involve the drainage system
within the eye. At the front of the eye there is a small space called the
anterior chamber. A clear fluid flows through this chamber and bathes and
nourishes the nearby tissues. (Watch the video to learn more about glaucoma. To
enlarge the video, click the brackets in the lower right-hand corner. To reduce
the video, press the Escape (Esc) button on your keyboard.) In glaucoma, for
still unknown reasons, the fluid drains too slowly out of the eye. As the fluid
builds up, the pressure inside the eye rises. Unless this pressure is
controlled, it may cause damage to the optic nerve and other parts of the eye
and result in loss of vision. Open-angle Glaucoma The most common type of
glaucoma is called open-angle glaucoma. In the normal eye, the clear fluid
leaves the anterior chamber at the open angle where the cornea and iris meet.
When fluid reaches the angle, it flows through a spongy meshwork, like a drain,
and leaves the eye. Sometimes, when the fluid reaches the angle, it passes too
slowly through the meshwork drain, causing the pressure inside the eye to build.
If the pressure damages the optic nerve, open-angle glaucoma -- and vision loss
-- may result. There is no cure for glaucoma. Vision lost from the disease
cannot be restored. However, there are treatments that may save remaining
vision. That is why early diagnosis is important.  See this graphic for a quick
overview of glaucoma,  including how many people it affects, whos at risk, what
to do if you have it, and how to learn more.  See a glossary of glaucoma terms.
Focus Area: Glaucoma
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--------------------
Example 1:
Question: What causes Glaucoma ?
Answer: Nearly 2.7 million people have glaucoma, a leading cause of blindness in
the United States. Although anyone can get glaucoma, some people are at higher
```

risk. They include - African-Americans over age 40 - everyone over age 60, especially Hispanics/Latinos - people with a family history of glaucoma. African-Americans over age 40 everyone over age 60, especially Hispanics/Latinos people with a family history of glaucoma. In addition to age, eye pressure is a risk factor. Whether you develop glaucoma depends on the level of pressure your optic nerve can tolerate without being damaged. This level is different for each person. Thats why a comprehensive dilated eye exam is very important. It can help your eye care professional determine what level of eye pressure is normal for you. Another risk factor for optic nerve damage relates to blood pressure. Thus, it is important to also make sure that your blood pressure is at a proper level for your body by working with your medical doctor. (Watch the animated video to learn more about the causes of glaucoma. To enlarge the video, click the brackets in the lower right-hand corner. To reduce the video, press the Escape (Esc) button on your keyboard.)
Focus Area: Glaucoma

Example 2:
Question: What are the symptoms of Glaucoma ?
Answer: Symptoms of Glaucoma Glaucoma can develop in one or both eyes. The most common type of glaucoma, open-angle glaucoma, has no symptoms at first. It causes no pain, and vision seems normal. Without treatment, people with glaucoma will slowly lose their peripheral, or side vision. They seem to be looking through a tunnel. Over time, straight-ahead vision may decrease until no vision remains. Tests for Glaucoma Glaucoma is detected through a comprehensive eye exam that includes a visual acuity test, visual field test, dilated eye exam, tonometry, and pachymetry. (Watch the animated video to learn more about testing for glaucoma. To enlarge the video, click the brackets in the lower right-hand corner. To reduce the video, press the Escape (Esc) button on your keyboard.) A visual acuity test uses an eye chart test to measure how well you see at various distances. A visual field test measures your side or peripheral vision. It helps your eye care professional tell if you have lost side vision, a sign of glaucoma. In a dilated eye exam, drops are placed in your eyes to widen, or dilate, the pupils. Your eye care professional uses a special magnifying lens to examine your retina and optic nerve for signs of damage and other eye problems. After the exam, your close-up vision may remain blurred for several hours. In tonometry, an instrument measures the pressure inside the eye. Numbing drops may be applied to your eye for this test. With pachymetry, a numbing drop is applied to your eye. Your eye care professional uses an ultrasonic wave instrument to measure the thickness of your cornea.
Focus Area: Glaucoma

Example 3:
Question: What are the treatments for Glaucoma ?
Answer: Although open-angle glaucoma cannot be cured, it can usually be controlled. While treatments may save remaining vision, they do not improve sight already lost from glaucoma. The most common treatments for glaucoma are

```
medication and surgery. Medications  Medications for glaucoma may be either in
the form of eye drops or pills. Some drugs reduce pressure by slowing the flow
of fluid into the eye. Others help to improve fluid drainage. (Watch the video
to learn more about coping with glaucoma. To enlarge the video, click the
brackets in the lower right-hand corner. To reduce the video, press the Escape
(Esc) button on your keyboard.) For most people with glaucoma, regular use of
medications will control the increased fluid pressure. But, these drugs may stop
working over time. Or, they may cause side effects. If a problem occurs, the eye
care professional may select other drugs, change the dose, or suggest other ways
to deal with the problem.  Read or listen to ways some patients are coping with
glaucoma. Surgery Laser surgery is another treatment for glaucoma. During laser
surgery, a strong beam of light is focused on the part of the anterior chamber
where the fluid leaves the eye. This results in a series of small changes that
makes it easier for fluid to exit the eye. Over time, the effect of laser
surgery may wear off. Patients who have this form of surgery may need to keep
taking glaucoma drugs. Researching Causes and Treatments Through studies in the
laboratory and with patients, NEI is seeking better ways to detect, treat, and
prevent vision loss in people with glaucoma. For example, researchers have
discovered genes that could help explain how glaucoma damages the eye. NEI also
is supporting studies to learn more about who is likely to get glaucoma, when to
treat people who have increased eye pressure, and which treatment to use first.
Focus Area: Glaucoma
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Example 4:
Question: What is (are) Glaucoma ?
Answer: Glaucoma is a group of diseases that can damage the eye's optic nerve
and result in vision loss and blindness. The most common form of the disease is
open-angle glaucoma. With early treatment, you can often protect your eyes
against serious vision loss. (Watch the video to learn more about glaucoma. To
enlarge the video, click the brackets in the lower right-hand corner. To reduce
the video, press the Escape (Esc) button on your keyboard.)  See this graphic
for a quick overview of glaucoma, including how many people it affects, whos at
risk, what to do if you have it, and how to learn more.  See a glossary of
glaucoma terms.
Focus Area: Glaucoma
--------------------------------------------------------------------------------
--------------------
Number of records: 14884
Number of unique focus_areas: 4985
Example of focus_areas: <ArrowStringArray>
[                          'Glaucoma',              'High Blood Pressure',
           'Paget's Disease of Bone',          'Urinary Tract Infections',
      'Alcohol Use and Older Adults',                    'Osteoarthritis',
              'Problems with Taste',                 'Anxiety Disorders',
                          'Diabetes',      'Medicare and Continuing Care',
                 'Knee Replacement',                  'Balance Problems',
 'Quitting Smoking for Older Adults',                   'Prostate Cancer',
```

```
                          'Dry Mouth',                        'Osteoporosis',
                     'Kidney Disease',                 'Alzheimer's Disease',
               'Rheumatoid Arthritis',                        'Hearing Loss',
                         'Low Vision',                                'COPD',
 'Age-related Macular Degeneration',                'Diabetic Retinopathy',
                         'Depression']
Length: 25, dtype: str
```

3 Dataset cleanup

- Even though the filtered data contain symptoms and the disease name (focus_area), the data is not clean. We make the text lowercase, remove leading and trailing spaces, remove URLs and remove cross reference sentences
- We create column for the actual embeddeding

[3]:
```
import re

def clean_text_rag(text):
    # If it's not string type
    if not isinstance(text, str):
        return ""

    text = text.lower()
    text = re.sub(r"\s+", " ", text)  # Replace weird spacing with normal space

    # Remove cross reference sentences
    text = re.sub(r"(Read more|More information)[^.]*\.", "", text, flags=re.
 ↪IGNORECASE)

    # Remove URLs
    text = re.sub(r"www\.\S+", "", text)

    return text.strip() # Remove leading and trailing white spaces

# Apply filters to each col
text_cols = filtered.select_dtypes(include="string").columns
for col in text_cols:
    filtered[col] = filtered[col].apply(clean_text_rag)

# Create newcolumn for embedding
filtered["text_for_rag"] = (
    "question: " + filtered["question"].fillna("") +
    " answer: " + filtered["answer"].fillna("")
)

display(filtered.head())
```

```
                                          question  \
```

```
0                 what is (are) glaucoma ?
1                   what causes glaucoma ?
2     what are the symptoms of glaucoma ?
3   what are the treatments for glaucoma ?
4                 what is (are) glaucoma ?

                                              answer           source  \
0  glaucoma is a group of diseases that can damag…  nihseniorhealth
1  nearly 2.7 million people have glaucoma, a lea…  nihseniorhealth
2  symptoms of glaucoma glaucoma can develop in o…  nihseniorhealth
3  although open-angle glaucoma cannot be cured, …  nihseniorhealth
4  glaucoma is a group of diseases that can damag…  nihseniorhealth

  focus_area                                       text_for_rag
0   glaucoma  question: what is (are) glaucoma ? answer: gla…
1   glaucoma  question: what causes glaucoma ? answer: nearl…
2   glaucoma  question: what are the symptoms of glaucoma ? …
3   glaucoma  question: what are the treatments for glaucoma…
4   glaucoma  question: what is (are) glaucoma ? answer: gla…
```

4 Create the embeddings

[4]:

```
from sentence_transformers import SentenceTransformer


embedder = SentenceTransformer("pritamdeka/S-PubMedBert-MS-MARCO")
embeddings = embedder.encode(filtered["text_for_rag"].tolist(),␣
 ↪show_progress_bar=True)

filtered["embedding"] = list(embeddings)
display(filtered.head())
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

Batches:   0%|          | 0/466 [00:00<?, ?it/s]

                                question  \
0                 what is (are) glaucoma ?
1                   what causes glaucoma ?
2     what are the symptoms of glaucoma ?
3   what are the treatments for glaucoma ?
4                 what is (are) glaucoma ?

                                              answer           source  \
0  glaucoma is a group of diseases that can damag…  nihseniorhealth
1  nearly 2.7 million people have glaucoma, a lea…  nihseniorhealth
```

```
2  symptoms of glaucoma glaucoma can develop in o…  nihseniorhealth
3  although open-angle glaucoma cannot be cured, …  nihseniorhealth
4  glaucoma is a group of diseases that can damag…  nihseniorhealth

  focus_area                                       text_for_rag  \
0   glaucoma  question: what is (are) glaucoma ? answer: gla…
1   glaucoma  question: what causes glaucoma ? answer: nearl…
2   glaucoma  question: what are the symptoms of glaucoma ? …
3   glaucoma  question: what are the treatments for glaucoma…
4   glaucoma  question: what is (are) glaucoma ? answer: gla…

                                           embedding
0  [-0.39189, -0.37175816, -0.44920766, -0.717546…
1  [-0.38811862, -0.65294784, -0.56563157, -0.618…
2  [-0.35303274, -0.45500723, -0.2737506, -0.8178…
3  [-0.47990558, -0.23519902, -0.37848306, -0.789…
4  [-0.3206859, -0.3648812, -0.4305763, -0.684870…
```

5 Create RAG

- We create a RAG with the embeddings and save it, so it can be later used in our pypeline

[5]:
```
import faiss
import numpy as np

embeddings = np.array(filtered["embedding"].tolist()).astype("float32")

index = faiss.IndexFlatL2(embeddings.shape[1])
index.add(embeddings)

# Save rag
import numpy as np
import pickle

faiss.write_index(index, ROOT + "rag/my_index.faiss")

# save embeddings
np.save(ROOT + "rag/embeddings.npy", embeddings)
```

6 Create query

- We create a RAG query that returns the top k best results. The results that are returned are already from the filtered dataset

[6]:
```
def query_rag(prompt, k = 5):
    query_vec = embedder.encode([prompt]).astype("float32")
    D, I = index.search(query_vec, k=k)
```

```
    results = filtered.iloc[I[0]]
    return results

# an example of the query
query_rag("Symptomps: Headache, feeling weak, sore throat, cough, really tired,␣
↪runny nose")
```

[6]:
```
                                               question  \
8219                     what are the symptoms of cough ?
8325   what are the symptoms of pleurisy and other pl…
910    what are the symptoms of small cell lung cancer ?
1577   do you have information about cold and cough m…
309                       what are the symptoms of copd ?

                                                 answer             source  \
8219   when you cough, mucus (a slimy substance) may …             nhlbi
8325   pleurisy the main symptom of pleurisy is a sha…             nhlbi
910    signs and symptoms of small cell lung cancer i…         cancergov
1577   summary : sneezing, sore throat, a stuffy nose…  mplushealthtopics
309    the most common symptoms of copd are a cough t…    nihseniorhealth

                                  focus_area  \
8219                                   cough
8325   pleurisy and other pleural disorders
910                   small cell lung cancer
1577                cold and cough medicines
309                                     copd

                                            text_for_rag  \
8219   question: what are the symptoms of cough ? ans…
8325   question: what are the symptoms of pleurisy an…
910    question: what are the symptoms of small cell …
1577   question: do you have information about cold a…
309    question: what are the symptoms of copd ? answ…

                                              embedding
8219   [-0.13024478, -0.48458353, -0.3863872, -0.4185…
8325   [-0.20189041, -0.42842674, -0.3009936, -0.7417…
910    [0.018616175, -0.5351676, -0.20493786, -0.7092…
1577   [-0.3706403, -0.28104895, -0.35472324, -0.6082…
309    [-0.031966053, -0.52303106, -0.23881392, -0.42…
```

6.1 Create Best Matching 25 and query for it

- We created a cleaning function that is pretty much the same as the one for the RAG, but on top of that includes removal of stop_words which should improve the performance
- We then build bm25 so we can query it

[7]:
```python
from rank_bm25 import BM25Okapi
import nltk

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

nltk.download("punkt")
nltk.download("stopwords")
nltk.download("punkt_tab")

stop_words = set(stopwords.words("english"))

def clean_text_BM25(text):
    # If it's not string type
    if not isinstance(text, str):
        return ""

    text = text.lower()
    text = re.sub(r"\s+", " ", text)  # Replace weird spacing with normal space

    # Remove cross reference sentences
    text = re.sub(r"(Read more|More information)[^.]*\.", "", text, flags=re.
↪IGNORECASE)

    # Remove URLs
    text = re.sub(r"www\.\S+", "", text)

    text = text.strip() # Remove leading and trailing white spaces

    # tokenize
    tokens = word_tokenize(text)

    # remove stopwords
    tokens = [
        word for word in tokens
        if word not in stop_words
    ]

    return " ".join(tokens)


# Create newcolumn for embedding
filtered["text_for_bm25"] = (filtered["answer"].apply(clean_text_BM25))
filtered.to_csv(ROOT + "dataset/medquad_filtered.csv")
display(filtered.head())
```

```python
# Build BM25 index on the same answer text
tokenized_corpus = [doc.lower().split() if isinstance(doc, str) else [] for doc␣
↪in filtered["text_for_bm25"].tolist()]
bm25 = BM25Okapi(tokenized_corpus)

# Save BM25
with open(ROOT + "bm25/bm25.pkl", "wb") as f:
    pickle.dump(bm25, f)
```

```
[nltk_data] Downloading package punkt to /home/sagemaker-
[nltk_data]     user/nltk_data…
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /home/sagemaker-
[nltk_data]     user/nltk_data…
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to /home/sagemaker-
[nltk_data]     user/nltk_data…
[nltk_data]   Package punkt_tab is already up-to-date!

                                 question  \
0                 what is (are) glaucoma ?
1                   what causes glaucoma ?
2     what are the symptoms of glaucoma ?
3  what are the treatments for glaucoma ?
4                 what is (are) glaucoma ?

                                              answer           source  \
0  glaucoma is a group of diseases that can damag…  nihseniorhealth
1  nearly 2.7 million people have glaucoma, a lea…  nihseniorhealth
2  symptoms of glaucoma glaucoma can develop in o…  nihseniorhealth
3  although open-angle glaucoma cannot be cured, …  nihseniorhealth
4  glaucoma is a group of diseases that can damag…  nihseniorhealth

  focus_area                                        text_for_rag  \
0   glaucoma  question: what is (are) glaucoma ? answer: gla…
1   glaucoma  question: what causes glaucoma ? answer: nearl…
2   glaucoma  question: what are the symptoms of glaucoma ? …
3   glaucoma  question: what are the treatments for glaucoma…
4   glaucoma  question: what is (are) glaucoma ? answer: gla…

                                           embedding  \
0  [-0.39189, -0.37175816, -0.44920766, -0.717546…
1  [-0.38811862, -0.65294784, -0.56563157, -0.618…
2  [-0.35303274, -0.45500723, -0.2737506, -0.8178…
3  [-0.47990558, -0.23519902, -0.37848306, -0.789…
4  [-0.3206859, -0.3648812, -0.4305763, -0.684870…

                                       text_for_bm25
```

```
0   glaucoma group diseases damage eye 's optic ne…
1   nearly 2.7 million people glaucoma , leading c…
2   symptoms glaucoma glaucoma develop one eyes . …
3   although open-angle glaucoma cured , usually c…
4   glaucoma group diseases damage eye 's optic ne…
```

7 Query for BM25

- Query for BM25 returns top k best results. The results are in a form of pandas dataframe from the filtered dataset

[8]:
```
def query_bm25(symptoms, k=5):
    # apply the same cleaning as corpus
    cleaned = clean_text_BM25(symptoms)
    tokens = cleaned.split()
    scores = bm25.get_scores(tokens)
    top_k = scores.argsort()[-k:][::-1]
    return filtered.iloc[top_k]

query_bm25("Headache, feeling weak, sore throat, cough, really tired, runny␣
 ↪nose")
```

[8]:
```
                                               question  \
1577  do you have information about cold and cough m…
8287             what are the symptoms of bronchitis ?
1602            what is (are) coronavirus infections ?
8219                  what are the symptoms of cough ?
2336                               what is (are) flu ?

                                                 answer             source  \
1577  summary : sneezing, sore throat, a stuffy nose…  mplushealthtopics
8287  acute bronchitis acute bronchitis caused by an…              nhlbi
1602  coronaviruses are common viruses that most peo…  mplushealthtopics
8219  when you cough, mucus (a slimy substance) may …              nhlbi
2336  flu is a respiratory infection caused by a num…  mplushealthtopics

                    focus_area  \
1577  cold and cough medicines
8287                bronchitis
1602    coronavirus infections
8219                     cough
2336                       flu

                                            text_for_rag  \
1577  question: do you have information about cold a…
8287  question: what are the symptoms of bronchitis …
1602  question: what is (are) coronavirus infections…
```

```
8219  question: what are the symptoms of cough ? ans…
2336  question: what is (are) flu ? answer: flu is a…

                                             embedding  \
1577  [-0.3706403, -0.28104895, -0.35472324, -0.6082…
8287  [-0.16960326, -0.5738766, -0.3613927, -0.28572…
1602  [-0.36878505, -0.47844404, -0.3749417, -0.5285…
8219  [-0.13024478, -0.48458353, -0.3863872, -0.4185…
2336  [-0.43634576, -0.5218197, -0.4388858, -0.46935…

                                         text_for_bm25
1577  summary : sneezing , sore throat , stuffy nose…
8287  acute bronchitis acute bronchitis caused infec…
1602  coronaviruses common viruses people get time l…
8219  cough , mucus ( slimy substance ) may come . c…
2336  flu respiratory infection caused number viruse…
```

8 Combine both queries

- We combine the both queries in order to get the top `m <= 2 * k` best results form RAG and BM25

[9]:
```python
def combined_query(symptoms, k=5):
    res_rag = query_rag(symptoms, k)
    res_bm25 = query_bm25(symptoms, k)

    # Combine and drop duplicate rows
    combined = pd.concat([res_rag, res_bm25]).drop_duplicates(subset="question")
    return combined

# Query example
combined_query("Headache, feeling weak, sore throat, cough, really tired, runny␣
 ↪nose")
```

[9]:
```
                                              question  \
8219                     what are the symptoms of cough ?
1577  do you have information about cold and cough m…
1718                      what is (are) sleep disorders ?
8325  what are the symptoms of pleurisy and other pl…
2251                          what is (are) sore throat ?
8287             what are the symptoms of bronchitis ?
1602            what is (are) coronavirus infections ?
2336                                  what is (are) flu ?

                                                answer             source  \
8219  when you cough, mucus (a slimy substance) may …              nhlbi
1577  summary : sneezing, sore throat, a stuffy nose…  mplushealthtopics
```

```
1718  is it hard for you to fall asleep or stay asle…  mplushealthtopics
8325  pleurisy the main symptom of pleurisy is a sha…              nhlbi
2251  your throat is a tube that carries food to you…  mplushealthtopics
8287  acute bronchitis acute bronchitis caused by an…              nhlbi
1602  coronaviruses are common viruses that most peo…  mplushealthtopics
2336  flu is a respiratory infection caused by a num…  mplushealthtopics

                               focus_area  \
8219                                cough
1577              cold and cough medicines
1718                       sleep disorders
8325  pleurisy and other pleural disorders
2251                          sore throat
8287                           bronchitis
1602                coronavirus infections
2336                                  flu

                                          text_for_rag  \
8219  question: what are the symptoms of cough ? ans…
1577  question: do you have information about cold a…
1718  question: what is (are) sleep disorders ? answ…
8325  question: what are the symptoms of pleurisy an…
2251  question: what is (are) sore throat ? answer: …
8287  question: what are the symptoms of bronchitis …
1602  question: what is (are) coronavirus infections…
2336  question: what is (are) flu ? answer: flu is a…

                                             embedding  \
8219  [-0.13024478, -0.48458353, -0.3863872, -0.4185…
1577  [-0.3706403, -0.28104895, -0.35472324, -0.6082…
1718  [-0.3134205, -0.47546673, -0.16874737, -0.5248…
8325  [-0.20189041, -0.42842674, -0.3009936, -0.7417…
2251  [-0.25978607, -0.5434519, -0.39540285, -0.4960…
8287  [-0.16960326, -0.5738766, -0.3613927, -0.28572…
1602  [-0.36878505, -0.47844404, -0.3749417, -0.5285…
2336  [-0.43634576, -0.5218197, -0.4388858, -0.46935…

                                         text_for_bm25
8219  cough , mucus ( slimy substance ) may come . c…
1577  summary : sneezing , sore throat , stuffy nose…
1718  hard fall asleep stay asleep night ? wake feel…
8325  pleurisy main symptom pleurisy sharp stabbing …
2251  throat tube carries food esophagus air windpip…
8287  acute bronchitis acute bronchitis caused infec…
1602  coronaviruses common viruses people get time l…
2336  flu respiratory infection caused number viruse…
```

9 Let's load dataset for the training the BERT

- We will use fine-tuned BERT on medical data to create a Classifier that will classify the medical urgency based on the symptoms into three categories: LOW/MEDIUM/HIGH.
- We use a two datasets for the training, each containing disease name and its symptoms
- First dataset contains 1065 records in format `description_of_symptoms` and `disease_name`. Datase can be found here link
- We use local LLM to extract only the symptoms from the description of the symptoms. Ollama with `llama3.2:3b` (or other model) is needed to be installed for this.
- The Classifier will then combine extracted user's symptoms with the best rag's symptoms and provide the urgency
- Even though we use a rather small dataset, the model should generalize since we feed it the symptoms and not the disease names. In pracitce, much larger dataset would be used to create this urgency classifier

[10]:
```
import argparse
import time
import ollama

from tqdm.auto import tqdm
from datasets import load_dataset

# Dataset one processing
dataset_one = load_dataset("gretelai/symptom_to_diagnosis")
dataset_one_train = dataset_one["train"].to_pandas()
dataset_one_test = dataset_one["test"].to_pandas()
dataset_one_full = pd.concat([dataset_one_train, dataset_one_test]).
    ↪reset_index(drop=True)
dataset_one_full.rename(columns={"output_text": "disease_name", "input_text":␣
    ↪"description_of_symptoms"}, inplace=True)

tqdm.pandas()

DEFAULT_MODEL = "llama3.2:3b"

EXTRACT_PROMPT = """Extract only the medical symptoms from the user's message.
Return a short comma-separated list of symptoms only. No explanation, no␣
    ↪preamble."""

def extract_symptoms(user_input: str) -> str:
    response = ollama.chat(
        model=DEFAULT_MODEL,
        messages=[
            {"role": "system", "content": EXTRACT_PROMPT},
            {"role": "user", "content": user_input},
        ],
    )
```

```
    return response.message.content.strip()


dataset_one_full["symptoms"] = dataset_one_full["description_of_symptoms"].
 ↪progress_map(extract_symptoms)
display(dataset_one_full.head())
print(dataset_one_full.shape)
```

```
  0%|          | 0/1065 [00:00<?, ?it/s]

             disease_name                            description_of_symptoms  \
0     cervical spondylosis  I've been having a lot of pain in my neck and …
1                 impetigo  I have a rash on my face that is getting worse…
2  urinary tract infection  I have been urinating blood. I sometimes feel …
3                arthritis  I have been having trouble with my muscles and…
4                   dengue  I have been feeling really sick. My body hurts…

                                            symptoms
0  Pain, balance problems, coordination issues, c…
1  Redness, inflammation, painful blisters with c…
2      Blood in urine, nausea upon urination, fever.
3  tension headaches, muscle weakness, swollen jo…
4  Body aches, loss of appetite, rashes on arms a…

(1065, 3)
```

9.1 Second dataset for BERT classifier

- For the second dataset we use this dataset that cotnains Disease name and its symptoms as a features.
- This dataset contains 4920 rows

[11]:
```
# Dataset two processing
dataset_two = pd.read_csv(ROOT + "dataset/DiseaseAndSymptoms.csv")
display(dataset_two.head())
# It contains columns with symptoms, we need to create one column that contains␣
 ↪all symptoms concatenated
# Combine all the symptoms that are not empty
dataset_two["symptoms"] = dataset_two.iloc[:, 1:].apply(
    lambda row: ",".join(row.dropna().astype(str).str.replace("_", " ")),
    axis=1
)
dataset_two.rename(columns={"Disease": "disease_name"}, inplace=True)
display(dataset_two.head())
print(dataset_two.shape[0])
```

```
           Disease   Symptom_1              Symptom_2              Symptom_3  \
0  Fungal infection    itching              skin_rash   nodal_skin_eruptions
1  Fungal infection  skin_rash   nodal_skin_eruptions    dischromic _patches
```

```
2  Fungal infection     itching   nodal_skin_eruptions    dischromic _patches
3  Fungal infection     itching              skin_rash    dischromic _patches
4  Fungal infection     itching              skin_rash   nodal_skin_eruptions

              Symptom_4 Symptom_5 Symptom_6 Symptom_7 Symptom_8 Symptom_9  \
0   dischromic _patches       NaN       NaN       NaN       NaN       NaN
1                   NaN       NaN       NaN       NaN       NaN       NaN
2                   NaN       NaN       NaN       NaN       NaN       NaN
3                   NaN       NaN       NaN       NaN       NaN       NaN
4                   NaN       NaN       NaN       NaN       NaN       NaN

  Symptom_10 Symptom_11 Symptom_12 Symptom_13 Symptom_14 Symptom_15  \
0        NaN        NaN        NaN        NaN        NaN        NaN
1        NaN        NaN        NaN        NaN        NaN        NaN
2        NaN        NaN        NaN        NaN        NaN        NaN
3        NaN        NaN        NaN        NaN        NaN        NaN
4        NaN        NaN        NaN        NaN        NaN        NaN

  Symptom_16 Symptom_17
0        NaN        NaN
1        NaN        NaN
2        NaN        NaN
3        NaN        NaN
4        NaN        NaN
       disease_name   Symptom_1              Symptom_2              Symptom_3  \
0  Fungal infection     itching              skin_rash   nodal_skin_eruptions
1  Fungal infection   skin_rash   nodal_skin_eruptions    dischromic _patches
2  Fungal infection     itching   nodal_skin_eruptions    dischromic _patches
3  Fungal infection     itching              skin_rash    dischromic _patches
4  Fungal infection     itching              skin_rash   nodal_skin_eruptions

              Symptom_4 Symptom_5 Symptom_6 Symptom_7 Symptom_8 Symptom_9  \
0   dischromic _patches       NaN       NaN       NaN       NaN       NaN
1                   NaN       NaN       NaN       NaN       NaN       NaN
2                   NaN       NaN       NaN       NaN       NaN       NaN
3                   NaN       NaN       NaN       NaN       NaN       NaN
4                   NaN       NaN       NaN       NaN       NaN       NaN

  Symptom_10 Symptom_11 Symptom_12 Symptom_13 Symptom_14 Symptom_15  \
0        NaN        NaN        NaN        NaN        NaN        NaN
1        NaN        NaN        NaN        NaN        NaN        NaN
2        NaN        NaN        NaN        NaN        NaN        NaN
3        NaN        NaN        NaN        NaN        NaN        NaN
4        NaN        NaN        NaN        NaN        NaN        NaN

  Symptom_16 Symptom_17                                          symptoms
0        NaN        NaN  itching, skin rash, nodal skin eruptions, disc…
```

```
1         NaN        NaN   skin rash, nodal skin eruptions, dischromic  …
2         NaN        NaN  itching, nodal skin eruptions, dischromic  pat…
3         NaN        NaN            itching, skin rash, dischromic  patches
4         NaN        NaN           itching, skin rash, nodal skin eruptions
```

```
4920
```

9.2 Combine the two datasets

- We need to combine both of the datasets and do cleanup, since some disease names are duplicated, just in a different format, so we unify them
- We are then left with 1356 unique symptoms and diseases. This will be used for the training

[12]:
```python
# Concatenate datset one and two to combine then
import pandas as pd

cols = ["disease_name", "symptoms"]

df_combined = pd.concat(
    [dataset_one_full[cols], dataset_two[cols]],
    ignore_index=True
)

#df_combined = dataset_one_full

display(df_combined.head())
print(df_combined.shape)
print(df_combined["disease_name"].nunique())
print(df_combined["disease_name"].unique())

# There are some duplicates, lets fix that and also some typos in the dataset
# Also in one dataset there is gerd and in the other "gastroesophageal reflux␣
↪disease", even tho its the same disease

alias_map = {
    "gerd": "gastroesophageal reflux disease",
    "peptic ulcer diseae": "peptic ulcer disease",
}
df_combined["disease_name"] = (
    df_combined["disease_name"]
    .str.strip()
    .str.lower()
    .str.replace("_", " ")
    .replace(alias_map)
)

print(f"Processed {50 * "-"}")
print(df_combined.shape)
```

```python
print(df_combined["disease_name"].nunique())
print(df_combined["disease_name"].unique())
print(df_combined["symptoms"].unique())
```

```
              disease_name                                           symptoms
0     cervical spondylosis  Pain, balance problems, coordination issues, c…
1                 impetigo  Redness, inflammation, painful blisters with c…
2  urinary tract infection      Blood in urine, nausea upon urination, fever.
3                arthritis  tension headaches, muscle weakness, swollen jo…
4                   dengue  Body aches, loss of appetite, rashes on arms a…

(5985, 2)
63
<ArrowStringArray>
[                   'cervical spondylosis',
                                'impetigo',
                 'urinary tract infection',
                               'arthritis',
                                  'dengue',
                             'common cold',
                           'drug reaction',
                        'fungal infection',
                                 'malaria',
                                 'allergy',
                        'bronchial asthma',
                          'varicose veins',
                                'migraine',
                            'hypertension',
         'gastroesophageal reflux disease',
                               'pneumonia',
                               'psoriasis',
                                'diabetes',
                                'jaundice',
                             'chicken pox',
                                 'typhoid',
                     'peptic ulcer disease',
                        'Fungal infection',
                                 'Allergy',
                                    'GERD',
                     'Chronic cholestasis',
                           'Drug Reaction',
                     'Peptic ulcer diseae',
                                    'AIDS',
                               'Diabetes ',
                         'Gastroenteritis',
                        'Bronchial Asthma',
                           'Hypertension ',
                                'Migraine',
```

```
                    'Cervical spondylosis',
            'Paralysis (brain hemorrhage)',
                                'Jaundice',
                                 'Malaria',
                             'Chicken pox',
                                  'Dengue',
                                 'Typhoid',
                             'hepatitis A',
                             'Hepatitis B',
                             'Hepatitis C',
                             'Hepatitis D',
                             'Hepatitis E',
                     'Alcoholic hepatitis',
                            'Tuberculosis',
                             'Common Cold',
                               'Pneumonia',
            'Dimorphic hemmorhoids(piles)',
                            'Heart attack',
                          'Varicose veins',
                          'Hypothyroidism',
                         'Hyperthyroidism',
                            'Hypoglycemia',
                         'Osteoarthristis',
                               'Arthritis',
 '(vertigo) Paroymsal  Positional Vertigo',
                                    'Acne',
                 'Urinary tract infection',
                               'Psoriasis',
                                'Impetigo']
Length: 63, dtype: str
Processed --------------------------------------------------
(5985, 2)
41
<ArrowStringArray>
[                   'cervical spondylosis',
                                'impetigo',
                 'urinary tract infection',
                               'arthritis',
                                  'dengue',
                             'common cold',
                           'drug reaction',
                        'fungal infection',
                                 'malaria',
                                 'allergy',
                        'bronchial asthma',
                          'varicose veins',
                                'migraine',
                            'hypertension',
```

```
          'gastroesophageal reflux disease',
                              'pneumonia',
                              'psoriasis',
                               'diabetes',
                               'jaundice',
                            'chicken pox',
                                'typhoid',
                     'peptic ulcer disease',
                      'chronic cholestasis',
                                   'aids',
                         'gastroenteritis',
             'paralysis (brain hemorrhage)',
                            'hepatitis a',
                            'hepatitis b',
                            'hepatitis c',
                            'hepatitis d',
                            'hepatitis e',
                      'alcoholic hepatitis',
                           'tuberculosis',
             'dimorphic hemmorhoids(piles)',
                           'heart attack',
                         'hypothyroidism',
                        'hyperthyroidism',
                           'hypoglycemia',
                        'osteoarthristis',
 '(vertigo) paroymsal  positional vertigo',
                                   'acne']
Length: 41, dtype: str
<ArrowStringArray>
[                   'Pain, balance problems, coordination issues, coughing, weak
limbs.',
                  'Redness, inflammation, painful blisters with clear pus,
facial rash.',
                                         'Blood in urine, nausea upon urination,
fever.',
       'tension headaches, muscle weakness, swollen joints, stiffness, pain in
the legs',
                      'Body aches, loss of appetite, rashes on arms and face,
eye pain.',
                                                  'Sore throat, coughing,
chills, fever',
                             'Rash, high fever, chills, headache, body aches,
eye pain.',
                                  'nausea, vomiting, fever, headache, weakness,
fatigue',
                              'redness, inflammation, dry skin, flakiness,
yellow fluid',
                                         'itching rash, flaky skin, dry skin
```

```
with marks',
 …
 ' skin rash, joint pain, silver like dusting, small dents in nails,
inflammatory nails',
        ' skin rash, joint pain, skin peeling, small dents in nails,
inflammatory nails',
         ' skin rash, joint pain, skin peeling, silver like dusting,
inflammatory nails',
       ' skin rash, joint pain, skin peeling, silver like dusting, small dents
in nails',
              ' skin rash, high fever, blister, red sore around nose, yellow
crust ooze',
                         ' high fever, blister, red sore around nose, yellow
crust ooze',
                          ' skin rash, blister, red sore around nose, yellow
crust ooze',
                       ' skin rash, high fever, red sore around nose, yellow
crust ooze',
                                   ' skin rash, high fever, blister, yellow
crust ooze',
                                ' skin rash, high fever, blister, red sore
around nose']
Length: 1357, dtype: str
```

10 Create urgency labels for dieseases

- We manually map urgencies labels LOW/MEDIUM/HIGH to the dieseases in the both datasets. The urgencies were mapped manually by severity index in here. Since this is only a proof of concept, it shouldn't matter if the dieses aren't labeled 100% correctly. In real world use this would be mapped by professionals in field.

[13]:
```
urgency_map = {
    # HIGH
    "heart attack": "HIGH",
    "paralysis (brain hemorrhage)": "HIGH",
    "pneumonia": "HIGH",
    "dengue": "HIGH",
    "malaria": "HIGH",
    "typhoid": "HIGH",
    "drug reaction": "HIGH",
    "hypoglycemia": "HIGH",
    "tuberculosis": "HIGH",
    "aids": "HIGH",
    "hepatitis b": "HIGH",
    "hepatitis c": "HIGH",
    "hepatitis d": "HIGH",
    "alcoholic hepatitis": "HIGH",
```

```
    # MEDIUM
    "urinary tract infection": "MEDIUM",
    "hypertension": "MEDIUM",
    "diabetes": "MEDIUM",
    "jaundice": "MEDIUM",
    "bronchial asthma": "MEDIUM",
    "peptic ulcer disease": "MEDIUM",
    "chicken pox": "MEDIUM",
    "chronic cholestasis": "MEDIUM",
    "gastroenteritis": "MEDIUM",
    "hepatitis a": "MEDIUM",
    "hepatitis e": "MEDIUM",
    "hyperthyroidism": "MEDIUM",
    "hypothyroidism": "MEDIUM",
    "dimorphic hemmorhoids(piles)": "MEDIUM",

    # LOW
    "cervical spondylosis": "LOW",
    "impetigo": "LOW",
    "arthritis": "LOW",
    "common cold": "LOW",
    "fungal infection": "LOW",
    "allergy": "LOW",
    "varicose veins": "LOW",
    "migraine": "LOW",
    "gastroesophageal reflux disease": "LOW",
    "psoriasis": "LOW",
    "osteoarthristis": "LOW",
    "(vertigo) paroymsal  positional vertigo": "LOW",
    "acne": "LOW",
}

def map_function(disease_name):
    return urgency_map[disease_name]


df_combined["urgency"] = df_combined["disease_name"].map(map_function)
display(df_combined.head())
```

```
             disease_name                                         symptoms  \
0     cervical spondylosis  Pain, balance problems, coordination issues, c…
1                 impetigo  Redness, inflammation, painful blisters with c…
2  urinary tract infection     Blood in urine, nausea upon urination, fever.
3                arthritis  tension headaches, muscle weakness, swollen jo…
4                   dengue  Body aches, loss of appetite, rashes on arms a…

  urgency
```

```
0      LOW
1      LOW
2   MEDIUM
3      LOW
4     HIGH
```

10.1 Gets combined query

- We create a combined query that will be used for both the actual pipeline and the BERT classifier

[14]:
```python
def first_sentence(text, max_chars=150):
    if not isinstance(text, str):
        return ""
    return text.split(".")[0].strip()[:max_chars]

# TODO: maybe extact the diesease names from the answer field in medquad␣
 ↪directly
def combined_query_processed(symptoms, k=3, max_length=200, disease_name =␣
 ↪True):
    res_rag = query_rag(symptoms, 4)
    res_bm25 = query_bm25(symptoms, 4)

    # Combine and drop duplicate rows
    combined = pd.concat([res_rag, res_bm25]).drop_duplicates(subset="question")
    focus_areas = []
    symptoms = []
    for _, row in combined.iterrows():
        focus_areas.append(row["focus_area"])
        symptoms.append(first_sentence(row["answer"], max_length)) # TODO:␣
 ↪answer can be replaced with text_for_rag or text_for_bm25

    string_result = []
    for i in range(len(combined)):
        if disease_name:
            string_result.append(f"Disease: {focus_areas[i]}, symptoms:␣
 ↪{symptoms[i]}.")
        else:
            string_result.append(f"symptoms: {symptoms[i]}.")

    return "".join(string_result)

res = combined_query_processed("Headache, feeling weak, sore throat, cough,␣
 ↪really tired, runny nose")
print(res)
```

```
Disease: cough, symptoms: when you cough, mucus (a slimy substance) may come
up.Disease: cold and cough medicines, symptoms: summary : sneezing, sore throat,
```

```
a stuffy nose, coughing -- everyone knows the symptoms of the common
cold.Disease: sleep disorders, symptoms: is it hard for you to fall asleep or
stay asleep through the night? do you wake up feeling tired or feel very sleepy
during the day, even if you have had enough sleep? you might have a sleep
disorder.Disease: pleurisy and other pleural disorders, symptoms: pleurisy the
main symptom of pleurisy is a sharp or stabbing pain in your chest that gets
worse when you breathe in deeply or cough or sneeze.Disease: bronchitis,
symptoms: acute bronchitis acute bronchitis caused by an infection usually
develops after you already have a cold or the flu.Disease: coronavirus
infections, symptoms: coronaviruses are common viruses that most people get some
time in their life.
```

- The final BERT classifier input will be the main symptoms along with the retrieved context from the RAG and BM25

[15]:
```
def create_bert_data(user_input):
    query = f"Main symptoms: {user_input} + Retrieved context:␣
 ↪{combined_query_processed(user_input, disease_name = False)}"
    return query

df_combined["bert_data"] = df_combined["symptoms"].
 ↪progress_map(create_bert_data)
display(df_combined.head())
```

```
  0%|          | 0/5985 [00:00<?, ?it/s]

              disease_name                                           symptoms  \
0     cervical spondylosis  Pain, balance problems, coordination issues, c…
1                 impetigo  Redness, inflammation, painful blisters with c…
2  urinary tract infection      Blood in urine, nausea upon urination, fever.
3                arthritis  tension headaches, muscle weakness, swollen jo…
4                   dengue  Body aches, loss of appetite, rashes on arms a…

  urgency                                          bert_data
0     LOW  Main symptoms: Pain, balance problems, coordin…
1     LOW  Main symptoms: Redness, inflammation, painful …
2  MEDIUM  Main symptoms: Blood in urine, nausea upon uri…
3     LOW  Main symptoms: tension headaches, muscle weakn…
4    HIGH  Main symptoms: Body aches, loss of appetite, r…
```

11 Prepare data for training the BioBert classifier

- Let's split the data into train, val, test. Splits will be 70%, 15%, 15%
- We also need to remove duplicates, since some combinations of symptoms show multiple times and this would create polution in the training. We are left with total of 1357 training data

[16]:
```
from sklearn.model_selection import train_test_split
```

```
dataset_one_symptoms = set(dataset_one_full["symptoms"].unique())
dataset_two_symptoms = set(dataset_two["symptoms"].unique())

print(f"Dataset one unique: {len(dataset_one_symptoms)}")
print(f"Dataset two unique: {len(dataset_two_symptoms)}")
print(f"Overlap between datasets: {len(dataset_one_symptoms &␣
 ↪dataset_two_symptoms)}")
print(f"Total unique combined: {len(dataset_one_symptoms |␣
 ↪dataset_two_symptoms)}")

# There are some overlaps in the symptoms, we only wanna keep unique ones, so␣
 ↪there is not data polution in the training
df_combined_dedup = df_combined.drop_duplicates(subset=["symptoms"]).
 ↪reset_index(drop=True)

# shuffle + split in one go
df_train, df_temp = train_test_split(
    df_combined_dedup,
    test_size=0.3,
    random_state=42,
    stratify=df_combined_dedup["urgency"]
)
df_val, df_test = train_test_split(
    df_temp,
    test_size=0.5,
    random_state=42,
    stratify=df_temp["urgency"]
)

print(df_train.shape)
print(df_val.shape)
print(df_test.shape)

# verify no overlap
overlap = pd.merge(df_train[["symptoms"]], df_val[["symptoms"]], on="symptoms")
print(f"Overlapping rows: {len(overlap)}")  # should be 0
```

```
Dataset one unique: 1053
Dataset two unique: 304
Overlap between datasets: 0
Total unique combined: 1357
(949, 4)
(204, 4)
(204, 4)
Overlapping rows: 0
```

11.1 Create a BIO-BERT classifier

- We use BERT classifier that was finetuned on medical data. The link to the classifier can be found here and more info about its architecture is here. In the report, we further explore its design
- We create train the model on 15 epochs, we use optimizer AdamW (default), learning_rate 2e-5 and weight_decay=0.01. We use an early stop mechanism that will stop the training if there is no improvement to the validation accuracy after 5 epochs.
- We also create a Metrics callback, that keeps track of entire training history

[18]:
```python
import logging
import torch
logging.getLogger("transformers").setLevel(logging.ERROR)
torch.cuda.empty_cache()
from transformers import AutoTokenizer

from transformers import AutoModelForSequenceClassification, TrainingArguments,␣
 ↪Trainer, EarlyStoppingCallback, TrainerCallback
from sklearn.metrics import f1_score, accuracy_score
import numpy as np

import torch

print(df_train.columns)
print(df_train["urgency"].unique())
print(df_train["urgency"].value_counts())

# Label mapping
label2id = {"LOW": 0, "MEDIUM": 1, "HIGH": 2}
id2label = {0: "LOW", 1: "MEDIUM", 2: "HIGH"}


df_train["label"] = df_train["urgency"].map(label2id)
df_val["label"]  = df_val["urgency"].map(label2id)
df_test["label"]  = df_test["urgency"].map(label2id)

tokenizer = AutoTokenizer.from_pretrained("dmis-lab/biobert-base-cased-v1.2")

# Create tokenizer
def tokenize(batch):
    return tokenizer(
        batch["bert_data"],
        truncation=True,
        padding="max_length",
        max_length=512,
    )

from datasets import Dataset
```

```
# Create datasets for the BERT
train_ds = Dataset.from_pandas(df_train[["bert_data", "label"]])
val_ds  = Dataset.from_pandas(df_val[["bert_data", "label"]])
test_ds  = Dataset.from_pandas(df_test[["bert_data", "label"]])

train_ds = train_ds.map(tokenize, batched=True)
val_ds  = val_ds.map(tokenize, batched=True)
test_ds  = test_ds.map(tokenize, batched=True)

train_ds.set_format("torch", columns=["input_ids", "attention_mask", "label"])
val_ds.set_format("torch", columns=["input_ids", "attention_mask", "label"])
test_ds.set_format("torch", columns=["input_ids", "attention_mask", "label"])

# Use pretraiend bio bert
model = AutoModelForSequenceClassification.from_pretrained(
    "dmis-lab/biobert-base-cased-v1.2",
    num_labels=3,
    id2label=id2label,
    label2id=label2id,
)

# Use gpu
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=-1)
    return {
        "accuracy": accuracy_score(labels, preds),
        "f1_macro": f1_score(labels, preds, average="macro"),
        "f1_weighted": f1_score(labels, preds, average="weighted"),
    }

# We wanna save the training history, more specifically train/val loss and␣
↪train/val acc
# we also print these metrics each epoch
class PrintMetricsCallback(TrainerCallback):
    def __init__(self):
        self.history = {
            "epoch": [],
            "train_loss": [],
            "val_loss": [],
            "val_accuracy": [],
            "val_f1_macro": [],
        }
```

```
        self._train_loss = None

    def on_log(self, args, state, control, logs=None, **kwargs):
        if logs and "loss" in logs and "eval_loss" not in logs:
            self._train_loss = logs["loss"]

    def on_evaluate(self, args, state, control, metrics=None, **kwargs):
        epoch = int(state.epoch)
        train_loss = self._train_loss or 0.0
        val_loss = metrics["eval_loss"]
        val_acc = metrics["eval_accuracy"]
        val_f1 = metrics["eval_f1_macro"]

        # store for plotting
        self.history["epoch"].append(epoch)
        self.history["train_loss"].append(train_loss)
        self.history["val_loss"].append(val_loss)
        self.history["val_accuracy"].append(val_acc)
        self.history["val_f1_macro"].append(val_f1)

        # print nicely
        print(
            f"Epoch {epoch:02d} | "
            f"train_loss: {train_loss:.4f} | "
            f"val_loss: {val_loss:.4f} | "
            f"val_acc: {val_acc:.4f} | "
            f"val_f1: {val_f1:.4f}"
        )

metrics_callback = PrintMetricsCallback()

# Define the training arguments
training_args = TrainingArguments(
    output_dir="./biobert-triage",
    num_train_epochs=15,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    eval_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    metric_for_best_model="f1_macro",
    logging_strategy="epoch",
    report_to="none",
    learning_rate=2e-5,
    weight_decay=0.01,
    warmup_ratio=0.1,
)
```

```python
# Create the training and train the model
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_ds,
    eval_dataset=val_ds,
    compute_metrics=compute_metrics,
    callbacks=[
        EarlyStoppingCallback(early_stopping_patience=5),
        metrics_callback,
    ],
)

trainer.train()

model.save_pretrained(ROOT + "models/biobert-triage-final")
tokenizer.save_pretrained(ROOT + "models/biobert-triage-final")
```

```
Index(['disease_name', 'symptoms', 'urgency', 'bert_data'], dtype='str')
<ArrowStringArray>
['MEDIUM', 'HIGH', 'LOW']
Length: 3, dtype: str
urgency
LOW       401
MEDIUM    304
HIGH      244
Name: count, dtype: int64

Map:   0%|          | 0/949 [00:00<?, ? examples/s]

Map:   0%|          | 0/204 [00:00<?, ? examples/s]

Map:   0%|          | 0/204 [00:00<?, ? examples/s]

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

{'loss': '1.037', 'grad_norm': '14.09', 'learning_rate': '1.318e-05', 'epoch':
'1'}

{'eval_loss': '0.868', 'eval_accuracy': '0.6814', 'eval_f1_macro': '0.5748',
'eval_f1_weighted': '0.6212', 'eval_runtime': '5.197',
'eval_samples_per_second': '39.26', 'eval_steps_per_second': '5.003', 'epoch':
'1'}
Epoch 01 | train_loss: 1.0370 | val_loss: 0.8680 | val_acc: 0.6814 | val_f1:
0.5748

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.7202', 'grad_norm': '13.68', 'learning_rate': '1.928e-05', 'epoch':
'2'}
```

```
{'eval_loss': '0.5338', 'eval_accuracy': '0.799', 'eval_f1_macro': '0.7839',
'eval_f1_weighted': '0.8012', 'eval_runtime': '5.206',
'eval_samples_per_second': '39.19', 'eval_steps_per_second': '4.994', 'epoch':
'2'}
Epoch 02 | train_loss: 0.7202 | val_loss: 0.5338 | val_acc: 0.7990 | val_f1:
0.7839

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.4003', 'grad_norm': '15.84', 'learning_rate': '1.78e-05', 'epoch':
'3'}

{'eval_loss': '0.4799', 'eval_accuracy': '0.8431', 'eval_f1_macro': '0.8149',
'eval_f1_weighted': '0.834', 'eval_runtime': '5.215', 'eval_samples_per_second':
'39.12', 'eval_steps_per_second': '4.986', 'epoch': '3'}
Epoch 03 | train_loss: 0.4003 | val_loss: 0.4799 | val_acc: 0.8431 | val_f1:
0.8149

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.2383', 'grad_norm': '19.81', 'learning_rate': '1.631e-05', 'epoch':
'4'}

{'eval_loss': '0.6312', 'eval_accuracy': '0.8235', 'eval_f1_macro': '0.7908',
'eval_f1_weighted': '0.8137', 'eval_runtime': '5.217',
'eval_samples_per_second': '39.1', 'eval_steps_per_second': '4.984', 'epoch':
'4'}
Epoch 04 | train_loss: 0.2383 | val_loss: 0.6312 | val_acc: 0.8235 | val_f1:
0.7908

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.1375', 'grad_norm': '1.683', 'learning_rate': '1.483e-05', 'epoch':
'5'}

{'eval_loss': '0.629', 'eval_accuracy': '0.848', 'eval_f1_macro': '0.8215',
'eval_f1_weighted': '0.8387', 'eval_runtime': '5.216',
'eval_samples_per_second': '39.11', 'eval_steps_per_second': '4.984', 'epoch':
'5'}
Epoch 05 | train_loss: 0.1375 | val_loss: 0.6290 | val_acc: 0.8480 | val_f1:
0.8215

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.0561', 'grad_norm': '0.3657', 'learning_rate': '1.335e-05', 'epoch':
'6'}

{'eval_loss': '0.5731', 'eval_accuracy': '0.8676', 'eval_f1_macro': '0.8476',
'eval_f1_weighted': '0.8624', 'eval_runtime': '5.223',
'eval_samples_per_second': '39.05', 'eval_steps_per_second': '4.978', 'epoch':
'6'}
Epoch 06 | train_loss: 0.0561 | val_loss: 0.5731 | val_acc: 0.8676 | val_f1:
0.8476
```

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.02447', 'grad_norm': '0.05722', 'learning_rate': '1.187e-05',
'epoch': '7'}

{'eval_loss': '0.546', 'eval_accuracy': '0.8922', 'eval_f1_macro': '0.8721',
'eval_f1_weighted': '0.8861', 'eval_runtime': '5.224',
'eval_samples_per_second': '39.05', 'eval_steps_per_second': '4.977', 'epoch':
'7'}
Epoch 07 | train_loss: 0.0245 | val_loss: 0.5460 | val_acc: 0.8922 | val_f1:
0.8721

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.004807', 'grad_norm': '6.444', 'learning_rate': '1.039e-05',
'epoch': '8'}

{'eval_loss': '0.6103', 'eval_accuracy': '0.8873', 'eval_f1_macro': '0.872',
'eval_f1_weighted': '0.8854', 'eval_runtime': '5.214',
'eval_samples_per_second': '39.12', 'eval_steps_per_second': '4.986', 'epoch':
'8'}
Epoch 08 | train_loss: 0.0048 | val_loss: 0.6103 | val_acc: 0.8873 | val_f1:
0.8720

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.003683', 'grad_norm': '0.0969', 'learning_rate': '8.904e-06',
'epoch': '9'}

{'eval_loss': '0.6431', 'eval_accuracy': '0.8873', 'eval_f1_macro': '0.8713',
'eval_f1_weighted': '0.8841', 'eval_runtime': '5.212',
'eval_samples_per_second': '39.14', 'eval_steps_per_second': '4.989', 'epoch':
'9'}
Epoch 09 | train_loss: 0.0037 | val_loss: 0.6431 | val_acc: 0.8873 | val_f1:
0.8713

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.0008957', 'grad_norm': '0.0324', 'learning_rate': '7.422e-06',
'epoch': '10'}

{'eval_loss': '0.6614', 'eval_accuracy': '0.8873', 'eval_f1_macro': '0.8716',
'eval_f1_weighted': '0.8848', 'eval_runtime': '5.212',
'eval_samples_per_second': '39.14', 'eval_steps_per_second': '4.989', 'epoch':
'10'}
Epoch 10 | train_loss: 0.0009 | val_loss: 0.6614 | val_acc: 0.8873 | val_f1:
0.8716

Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]

{'loss': '0.0005202', 'grad_norm': '0.1802', 'learning_rate': '5.94e-06',
'epoch': '11'}

{'eval_loss': '0.6881', 'eval_accuracy': '0.8775', 'eval_f1_macro': '0.8609',
'eval_f1_weighted': '0.8747', 'eval_runtime': '5.208',
```

```
'eval_samples_per_second': '39.17', 'eval_steps_per_second': '4.993', 'epoch':
'11'}
Epoch 11 | train_loss: 0.0005 | val_loss: 0.6881 | val_acc: 0.8775 | val_f1:
0.8609
```

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

```
{'loss': '0.0004338', 'grad_norm': '0.02525', 'learning_rate': '4.458e-06',
'epoch': '12'}
```

```
{'eval_loss': '0.6956', 'eval_accuracy': '0.8725', 'eval_f1_macro': '0.8544',
'eval_f1_weighted': '0.8692', 'eval_runtime': '5.214',
'eval_samples_per_second': '39.12', 'eval_steps_per_second': '4.986', 'epoch':
'12'}
Epoch 12 | train_loss: 0.0004 | val_loss: 0.6956 | val_acc: 0.8725 | val_f1:
0.8544
```

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

```
{'train_runtime': '1042', 'train_samples_per_second': '13.66',
'train_steps_per_second': '1.713', 'train_loss': '0.2187', 'epoch': '12'}
```

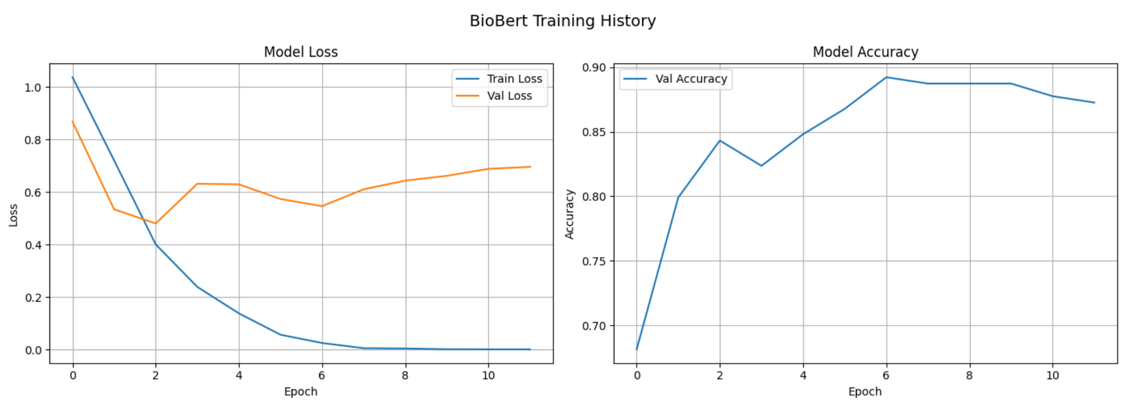
```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

[18]:
```
('../models/biobert-triage-final/tokenizer_config.json',
 '../models/biobert-triage-final/tokenizer.json')
```

11.2 Visualise the training

[22]:
```python
import matplotlib.pyplot as plt
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Accuracy plot
ax1.plot(metrics_callback.history["train_loss"], label='Train Loss')
ax1.plot(metrics_callback.history["val_loss"], label='Val Loss')
ax1.set_title('Model Loss')
ax1.set_xlabel('Epoch')
ax1.set_ylabel('Loss')
ax1.legend()
ax1.grid(True)

# Loss plot
ax2.plot(metrics_callback.history["val_accuracy"], label='Val Accuracy')
ax2.set_title('Model Accuracy')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Accuracy')
ax2.legend()
ax2.grid(True)

plt.suptitle('BioBert Training History', fontsize=14)
```

```python
plt.tight_layout()
plt.savefig('training_history_biobert.png', dpi=150, bbox_inches='tight')
plt.show()


train_metrics = trainer.evaluate(train_ds)
print(f"Train loss: {train_metrics['eval_loss']:.4f}")
print(f"Train accuracy: {train_metrics['eval_accuracy']:.4f}")

val_metrics = trainer.evaluate(val_ds)
print(f"Val loss: {val_metrics['eval_loss']:.4f}")
print(f"Val accuracy: {val_metrics['eval_accuracy']:.4f}")

test_metrics = trainer.evaluate(test_ds)
print(f"Test loss: {test_metrics['eval_loss']:.4f}")
print(f"Test accuracy: {test_metrics['eval_accuracy']:.4f}")
```

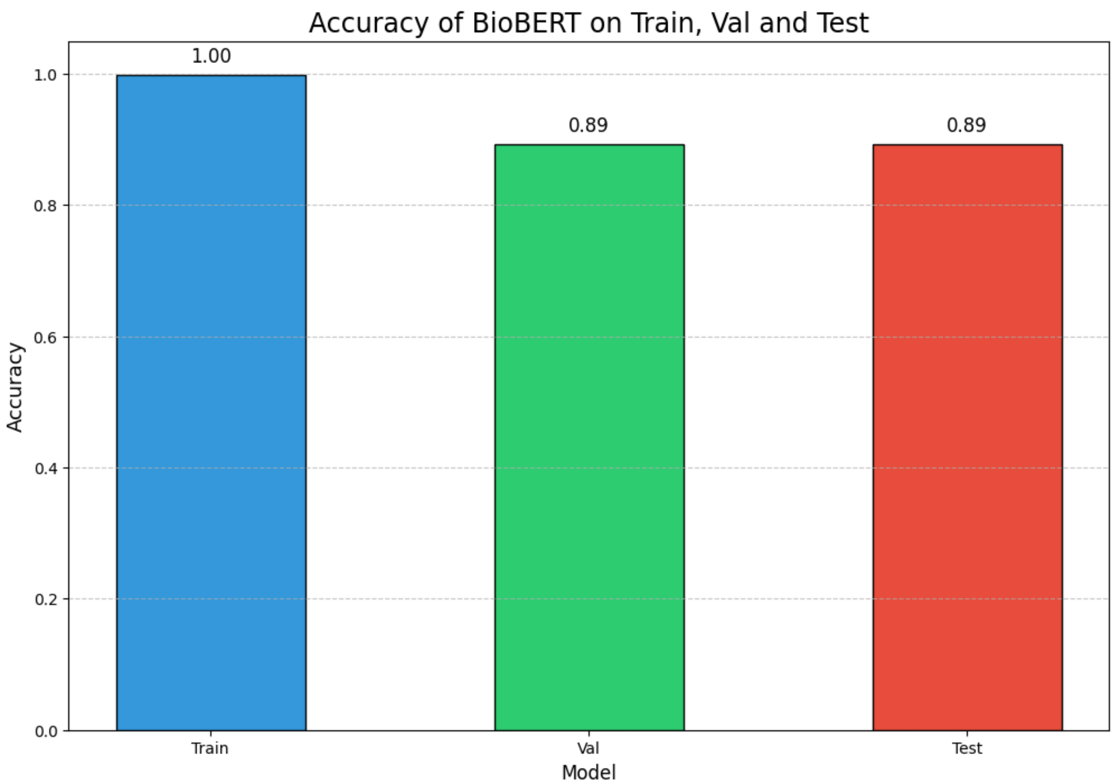



```
{'eval_loss': '0.003385', 'eval_accuracy': '0.9989', 'eval_f1_macro': '0.9988',
'eval_f1_weighted': '0.9989', 'eval_runtime': '23.75',
'eval_samples_per_second': '39.96', 'eval_steps_per_second': '5.01', 'epoch':
'12'}
Epoch 12 | train_loss: 0.0004 | val_loss: 0.0034 | val_acc: 0.9989 | val_f1:
0.9988
Train loss: 0.0034
Train accuracy: 0.9989

{'eval_loss': '0.546', 'eval_accuracy': '0.8922', 'eval_f1_macro': '0.8721',
'eval_f1_weighted': '0.8861', 'eval_runtime': '5.16', 'eval_samples_per_second':
'39.54', 'eval_steps_per_second': '5.039', 'epoch': '12'}
Epoch 12 | train_loss: 0.0004 | val_loss: 0.5460 | val_acc: 0.8922 | val_f1:
0.8721
Val loss: 0.5460
Val accuracy: 0.8922
```

```
{'eval_loss': '0.5809', 'eval_accuracy': '0.8922', 'eval_f1_macro': '0.8847',
'eval_f1_weighted': '0.8905', 'eval_runtime': '5.156',
'eval_samples_per_second': '39.57', 'eval_steps_per_second': '5.043', 'epoch':
'12'}
Epoch 12 | train_loss: 0.0004 | val_loss: 0.5809 | val_acc: 0.8922 | val_f1:
0.8847
Test loss: 0.5809
Test accuracy: 0.8922
```

[33]:

```python
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(12,8))

colors = ['#3498db', '#2ecc71', '#e74c3c']

labels = ['Train', 'Val', 'Test']
values = [train_metrics['eval_accuracy'] ,val_metrics['eval_accuracy'] ,␣
↪test_metrics['eval_accuracy']]

ax.bar(labels, values, color=colors, edgecolor='black', width=0.5)

ax.set_ylim(0, 1.05)
ax.set_yticks([0, 0.2, 0.4, 0.6, 0.8, 1.0])

ax.set_xlabel("Model", fontsize=12)
ax.set_ylabel("Accuracy", fontsize=13)
ax.set_title("Accuracy of BioBERT on Train, Val and Test", fontsize=17)

ax.grid(axis='y', linestyle='--', alpha=0.7)

for i, value in enumerate(values):
    ax.text(i, value + 0.02, f'{value:.2f}', ha='center', fontsize=12)

plt.show()
```



11.3 Inference

- Testing the biobert

[31]:
```python
def predict_urgency(user_symptoms):
    enriched = create_bert_data(user_symptoms)
    inputs = tokenizer(
        enriched,
        return_tensors="pt",
        truncation=True,
        max_length=512
    )
    # move inputs to same device as model
    inputs = {k: v.to(device) for k, v in inputs.items()}

    with torch.no_grad():
        outputs = model(**inputs)
    pred = outputs.logits.argmax(-1).item()
    return id2label[pred]


# Test it
```

```python
# Low urgency - basic conditions
print(f"Low urgency: {"-" * 50}")
print(predict_urgency("runny nose, mild cough, sneezing"))
print(predict_urgency("slight itching, dry skin, mild rash"))
print(predict_urgency("mild headache, feeling tired"))

# Medium urgency
print(f"Medium urgency: {"-" * 50}")
print(predict_urgency("yellowing of skin and eyes, dark urine, fatigue"))  #␣
↪jaundice
print(predict_urgency("wheezing, shortness of breath, chest tightness"))  #␣
↪asthma attack

# High urgency
print(f"High urgency: {"-" * 50}")
print(predict_urgency("coughing up blood, difficulty breathing, high fever,␣
↪chills")) # Severe pneumonia
print(predict_urgency("sudden severe chest pain radiating to left arm,␣
↪sweating, difficulty breathing")) # Heart attack
print(predict_urgency("severe abdominal pain, vomiting blood, rapid heartbeat,␣
↪dizziness")) # Internal bleeding
print(predict_urgency("sudden weakness on one side of body, slurred speech,␣
↪drooping face")) # Stroke
```

```
Low urgency: --------------------------------------------------
LOW
MEDIUM

MEDIUM
Medium urgency: --------------------------------------------------
MEDIUM
LOW
High urgency: --------------------------------------------------

HIGH
HIGH
HIGH

LOW
```

[ ]:

```

```

[ ]:

```

```